



Submitted to:-

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Submitted by:-

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Reg no:-

2023-BSE-021

Section:-

5 A

Cloud Computing

Lab Exam

<https://github.com/Hafsa-Khalid021/Lab-Exam>

Q1 – AWS IAM Setup Using AWS CLI and Console Verification

```
● @Hafsa-Khalid021 →/workspaces/Lab-Exam (main) $ aws iam create-group --group-name SoftwareEngineering
{
  "Group": {
    "Path": "/",
    "GroupName": "SoftwareEngineering",
    "GroupId": "AGPA3ED5IU4DHJRECLPKS",
    "Arn": "arn:aws:iam::764766889734:group/SoftwareEngineering",
    "CreateDate": "2026-01-19T07:43:25+00:00"
  }
}
○ @Hafsa-Khalid021 →/workspaces/Lab-Exam (main) $ █

● @Hafsa-Khalid021 →/workspaces/Lab-Exam (main) $ aws iam get-group --group-name SoftwareEngineering
{
  "Users": [],
  "Group": {
    "Path": "/",
    "GroupName": "SoftwareEngineering",
    "GroupId": "AGPA3ED5IU4DHJRECLPKS",
    "Arn": "arn:aws:iam::764766889734:group/SoftwareEngineering",
    "CreateDate": "2026-01-19T07:43:25+00:00"
  }
}
○ @Hafsa-Khalid021 →/workspaces/Lab-Exam (main) $ █

● @Hafsa-Khalid021 →/workspaces/Lab-Exam (main) $ aws iam create-user --user-name Hafsa
{
  "User": {
    "Path": "/",
    "UserName": "Hafsa",
    "UserId": "AIDA3ED5IU4DNWZ3X25HI",
    "Arn": "arn:aws:iam::764766889734:user/Hafsa",
    "CreateDate": "2026-01-19T07:45:00+00:00"
  }
}

● @Hafsa-Khalid021 →/workspaces/Lab-Exam (main) $ aws iam add-user-to-group \
  --user-name Hafsa \
  --group-name SoftwareEngineering
○ @Hafsa-Khalid021 →/workspaces/Lab-Exam (main) $ █
```

```
● @Hafsa-Khalid021 → /workspaces/Lab-Exam (main) $ aws iam get-group --group-name SoftwareEngineering
{
  "Users": [
    {
      "Path": "/",
      "UserName": "Hafsa",
      "UserId": "AIDA3ED5IU4DNWZ3X25HI",
      "Arn": "arn:aws:iam::764766889734:user/Hafsa",
      "CreateDate": "2026-01-19T07:45:00+00:00"
    }
  ],
  "Group": {
    "Path": "/",
    "GroupName": "SoftwareEngineering",
    "GroupId": "AGPA3ED5IU4DHJRECLPKS",
    "Arn": "arn:aws:iam::764766889734:group/SoftwareEngineering",
    "CreateDate": "2026-01-19T07:43:25+00:00"
  }
}
```

```
○ @Hafsa-Khalid021 → /workspaces/Lab-Exam (main) $
```

```
● @Hafsa-Khalid021 → /workspaces/Lab-Exam (main) $ aws iam list-policies --scope AWS --query "Policies[?PolicyName=='AdministratorAccess']"
[
  {
    "PolicyName": "AdministratorAccess",
    "PolicyId": "ANPAIWMBCKSKIEE64ZLYK",
    "Arn": "arn:aws:iam::aws:policy/AdministratorAccess",
    "Path": "/",
    "DefaultVersionId": "v1",
    "AttachmentCount": 0,
    "PermissionsBoundaryUsageCount": 0,
    "IsAttachable": true,
    "CreateDate": "2015-02-06T18:39:46+00:00",
    "UpdateDate": "2015-02-06T18:39:46+00:00"
  }
]
```

```
● @Hafsa-Khalid021 → /workspaces/Lab-Exam (main) $ aws iam attach-group-policy \
--group-name SoftwareEngineering \
--policy-arn arn:aws:iam::aws:policy/AdministratorAccess
```

```
● @Hafsa-Khalid021 → /workspaces/Lab-Exam (main) $ aws iam list-attached-group-policies \
--group-name SoftwareEngineering
{
  "AttachedPolicies": [
    {
      "PolicyName": "AdministratorAccess",
      "PolicyArn": "arn:aws:iam::aws:policy/AdministratorAccess"
    }
  ]
}
```

```
○ @Hafsa-Khalid021 → /workspaces/Lab-Exam (main) $
```

The screenshot shows the AWS IAM console interface. At the top, there's a navigation bar with the AWS logo, a search bar, and user information for 'Hafsa Khalid (7647-6688-9734)'. The main content area is divided into two sections: 'User groups (1)' and 'Users (1)'. The 'User groups' section shows a table with one group named 'SoftwareEngineering'. The 'Users' section shows a table with one user named 'Hafsa'.

User groups (1)

A user group is a collection of IAM users. Use groups to specify permissions for a collection of users.

Group name	Users	Permissions	Creation time
SoftwareEngineering	0	Not defined	Now

Users (1)

An IAM user is an identity with long-term credentials that is used to interact with AWS in an account.

User name	Path	Group	Last activity	MFA	Password age	Console
Hafsa	/	1	-	-	-	-

Q2 – Terraform Lab: Simple AWS Environment with Nginx over HTTPS

```
@Hafsa-Khalid021 → /workspaces/Lab-Exam (main) $ terraform init
Initializing the backend...
Initializing provider plugins...
- Finding latest version of hashicorp/http...
- Reusing previous version of hashicorp/aws from the dependency lock file
- Installing hashicorp/http v3.5.0...
- Installed hashicorp/http v3.5.0 (signed by HashiCorp)
- Using previously-installed hashicorp/aws v6.28.0
Terraform has made some changes to the provider dependency selections recorded
in the .terraform.lock.hcl file. Review those changes and commit them to your
version control system if they represent changes you intended to make.
```

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see any changes that are required for your infrastructure. All Terraform commands should now work.

If you ever set or change modules or backend configuration for Terraform, rerun this command to reinitialize your working directory. If you forget, other commands will detect it and remind you to do so if necessary.

```
@Hafsa-Khalid021 →/workspaces/Lab-Exam (main) $ terraform plan
```

```
+ ipv6_cidr_block          = (known after apply)
+ ipv6_cidr_block_network_border_group = (known after apply)
+ main_route_table_id      = (known after apply)
+ owner_id                 = (known after apply)
+ region                   = "me-central-1"
+ tags                     = {
  + "Name" = "dev-vpc"
}
+ tags_all                 = {
  + "Name" = "dev-vpc"
}
}
```

Plan: 7 to add, 0 to change, 0 to destroy.

Changes to Outputs:

```
+ ec2_public_ip = (known after apply)
```

Note: You didn't use the -out option to save this plan, so Terraform can't guarantee to take exactly these actions if you run "terraform apply" now.

```
@Hafsa-Khalid021 →/workspaces/Lab-Exam (main) $
```

```
@Hafsa-Khalid021 →/workspaces/Lab-Exam (main) $ terraform apply
```

```
}
```

Plan: 1 to add, 0 to change, 0 to destroy.

Changes to Outputs:

```
+ ec2_public_ip = (known after apply)
```

Do you want to perform these actions?

Terraform will perform the actions described above.

Only 'yes' will be accepted to approve.

Enter a value: yes

aws_instance.myapp_ec2: Creating...

aws_instance.myapp_ec2: Still creating... [00m10s elapsed]

aws_instance.myapp_ec2: Creation complete after 14s [id=i-07b68bf175df7e9b2]

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.

Outputs:

```
ec2_public_ip = "51.112.50.128"
```

```
@Hafsa-Khalid021 →/workspaces/Lab-Exam (main) $
```

- @Hafsa-Khalid021 → /workspaces/Lab-Exam (main) \$ terraform output
ec2_public_ip = "51.112.50.128"
- @Hafsa-Khalid021 → /workspaces/Lab-Exam (main) \$

aws [Search] [Alt+S] United States (N. Virginia) Hafsa Khalid (7647-6688-9734) Hafsa%20Khalid

VPC > Your VPCs

VPC dashboard < AWS Global View Filter by VPC

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

Your VPCs

VPCs VPC encryption controls

Your VPCs (2) Info Last updated less than a minute ago Actions Create VPC

Find VPCs by attribute or tag

☐	Name	VPC ID	State	Encryption c...	Encryption control ...	Blc
☐	lab-vpc	vpc-0888b1d106aa0d99a	Available	-	-	
☐	-	vpc-04a2092ce7da55131	Available	-	-	

aws [Search] [Alt+S] United States (N. Virginia) Hafsa Khalid (7647-6688-9734) Hafsa%20Khalid

VPC > Subnets

VPC dashboard < AWS Global View Filter by VPC

Virtual private cloud

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Subnets

Route tables

Internet gateways

Egress-only internet gateways

Carrier gateways

Subnets

Subnets (1/8) Info Last updated 1 minute ago Actions Create subnet

Find subnets by attribute or tag

☒	Name	Subnet ID	State	VPC	Block Publi
☒	-	subnet-024d513e21746244b	Available	vpc-04a2092ce7da55131	Off
☐	-	subnet-050bc5e466d31bea4	Available	vpc-04a2092ce7da55131	Off
☐	-	subnet-0ac81195056dd8fbc	Available	vpc-0888b1d106aa0d99a	Off
☐	-	subnet-07908eea82eb1aa23	Available	vpc-04a2092ce7da55131	Off

subnet-024d513e21746244b

Details Flow logs Route table Network ACL CIDR reservations Sharing Tags

aws [Search] [Alt+S] United States (N. Virginia) Hafsa Khalid (7647-6688-9734) Hafsa%20Khalid

VPC > Internet gateways

VPC dashboard < AWS Global View Filter by VPC

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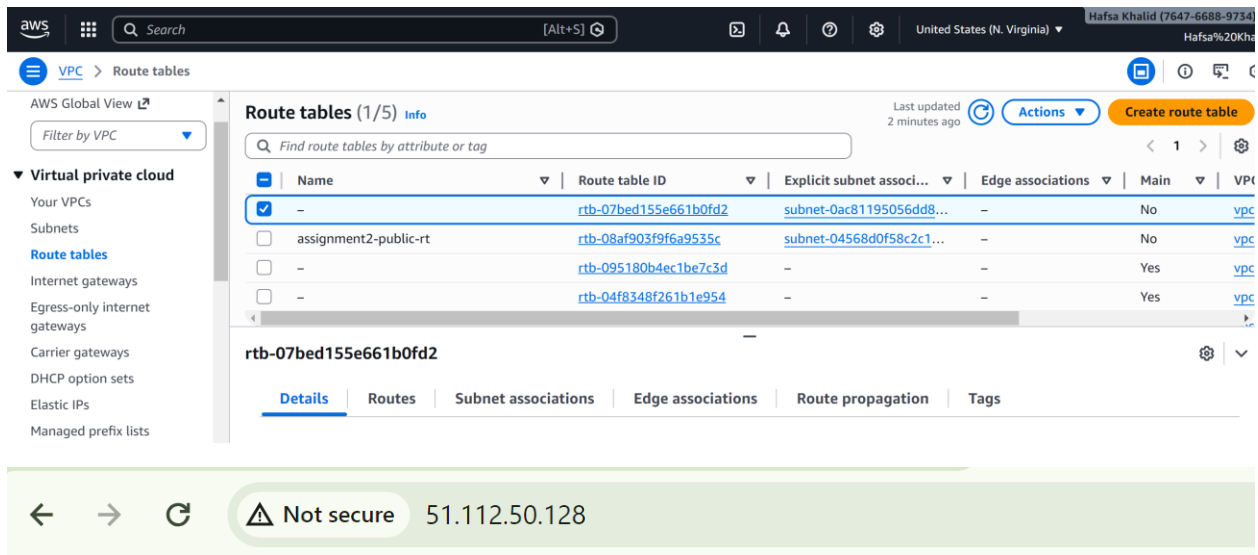
Route tables

Internet gateways

Internet gateways (2) Info Actions Create internet gateway

Find internet gateways by attribute or tag

☐	Name	Internet gateway ID	State	VPC ID
☐	-	igw-0ad9c6ca7f3f6f68c	Attached	vpc-0888b1d106aa0d99a
☐	-	igw-0be30876333b2eedf	Attached	vpc-04a2092ce7da55131



This is Hafsa's Terraform environment

Q3 – Ansible Playbook for EC2 Web Server Using Q2 Instance

```
@Hafsa-Khalid021 → /workspaces/Lab-Exam (main) $ ansible-playbook -i hosts nginx.yml --user ec2-user --private-key ~/.ssh/id_ed25519

PLAY [Install nginx on EC2] *****

TASK [Gathering Facts] *****
[WARNING]: Host '51.112.50.128' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.20/reference_appendices/interpreter_discovery.html for more information.
ok: [51.112.50.128]

TASK [Install nginx] *****
ok: [51.112.50.128]

TASK [Start nginx] *****
ok: [51.112.50.128]

PLAY RECAP *****
51.112.50.128 : ok=3 changed=0 unreachable=0 failed=0 skipped=0 rescued=0 ignored=0

@Hafsa-Khalid021 → /workspaces/Lab-Exam (main) $
```

← → ↻ ⚠ Not secure 51.112.50.128

This is Hafsa's Terraform environment